

TEMPEST MAINFRAME



QUARTIC PULSE

Audio-Driven Modulation of Quartic Complex Fractals and a Ray-Marched Power-4 Mandelbulb

Technical Methods Paper

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This paper documents the implemented mathematics and signal-processing path in the v0.27 renderer. It is an engineering methods report, not a claim of peer review. Equations and constants are transcribed from the working WebGL2 and Web Audio implementation so the behavior can be studied, reproduced, and improved.

Human-directed, AI-assisted open-source development

Abstract

Quartic Pulse is a real-time music visualizer in which extracted audio features alter the parameters of fractal recurrences, not only the color or position of a precomputed image. The two-dimensional signature visual uses the perturbed quartic recurrence $z_{(k+1)} = z_k^4 + \mu(t) z_k^2 + c'$, with optional rotating fold and nonlinear warp operators. Version 0.27 adds a three-dimensional power-4 Mandelbulb evaluated as a distance field and displayed by adaptive sphere tracing on the GPU. Audio is sampled through the Web Audio API, separated into low, mid, and high frequency bands, normalized with asymmetric automatic gain control, smoothed, and converted into bass, mid, high, RMS, beat, and dominant-frequency features. Those features control the exponent, folding, camera, palette position, surface lighting, and general user-defined modulation routes. This paper gives the implemented equations, coefficient values, rendering method, performance policy, validation results, and limitations needed to reproduce the v0.27 behavior.

Keywords: audio-reactive graphics; Mandelbrot set; Mandelbulb; quartic recurrence; WebGL2; FFT; distance estimation; sphere tracing; generative art

1 Introduction

Many music visualizers treat an image generator and an audio meter as separate systems: the image is created first, and sound later changes brightness, scale, or color. Quartic Pulse uses a stronger coupling. Its audio features are supplied to the recurrence and distance-estimator parameters before the GPU evaluates each pixel. Consequently, the boundary, lobes, orbit traps, folds, and three-dimensional surface can change with the music.

The design objective is expressive motion without losing the deterministic structure that makes a fractal recognizable. A second objective is accessibility on an average Windows 11 computer. Live rendering therefore uses bounded iteration counts and adaptive resolution, while offline export is allowed a larger computation budget.

The project is open source under GPL-3.0-or-later. The methods described here correspond to Quartic Pulse v0.27.0 and should be understood as a documented software implementation rather than a proposed canonical definition of a three-dimensional Mandelbrot set.

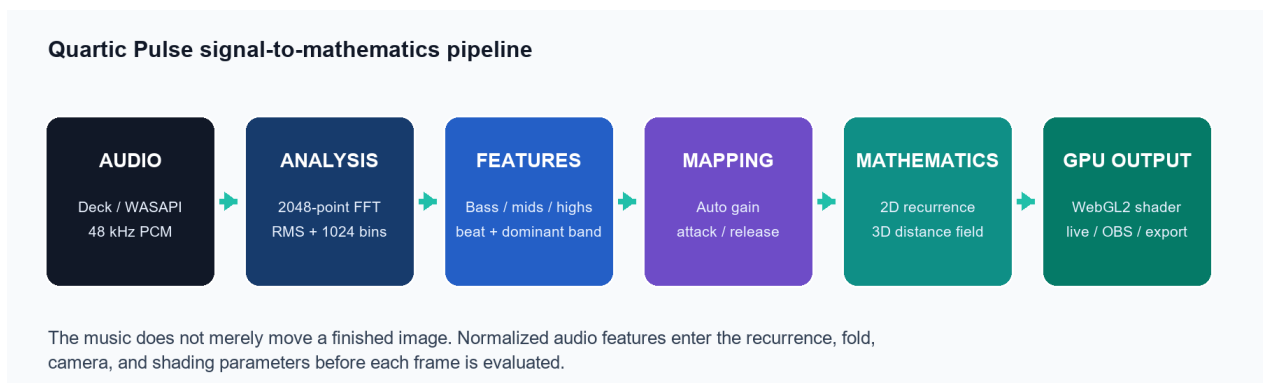


Figure 1. Signal-to-mathematics pipeline implemented in Quartic Pulse v0.27.

2 System Architecture

The desktop program is an Electron application. The renderer uses WebGL2 fragment shaders for the visual field and the Web Audio API for frequency- and time-domain measurements. Loaded tracks, Windows playback endpoints, and compatible input or virtual devices enter a common analysis graph. A native WASAPI helper can

supply PCM samples from Windows endpoints that are not directly exposed as ordinary browser capture devices.

The CPU-side analysis produces compact, normalized state values in the interval [0,1]. Each frame transfers these values as shader uniforms. The two-dimensional fractal, the three-dimensional distance field, the conventional spectrum styles, OBS output, and video exporter therefore share the same musical state. During offline export, visual time is tied to audio time rather than wall-clock frame time, improving repeatability.

Table 1. Major processing stages.

Stage	Implemented mechanism	Output
Acquisition	MediaElement, Web Audio input, or WASAPI PCM worklet	48 kHz analysis stream
Spectral analysis	AnalyserNode, FFT size 2048, byte frequency data	1024 magnitude bins
Feature extraction	Band means, RMS, beat detector, dominant-band estimator	B, M, H, R, Q, hue
Control mapping	Direct shader formulas plus configurable attack/release routes	Bounded modulation values
Rendering	WebGL2 escape-time iteration or distance-estimated ray marching	Live, OBS, or exported frames

3 Two-Dimensional Quartic Fractal

3.1 Base recurrence

Let z_k and c be complex numbers represented on the GPU by two-component vectors. Complex multiplication is implemented explicitly, so z^4 is evaluated as $(z^2)^2$. For the signature Quartic Mandelbrot option, $z_0 = 0$ and the iteration is:

EQUATION 1 - IMPLEMENTED QUARTIC RECURRENCE

$$z_{(k+1)} = z_k^4 + \mu(t) z_k^2 + c'$$

With $\mu(t) = 0$, this reduces to the ordinary power-4 escape-time recurrence.

The coordinate term is $c' = s_c c + b$, where b is a user-set complex bias and s_c linearly maps the core strength control from 0.35 to 1.65. The default core strength is 0.5, producing $s_c = 1.0$. Escape is declared when $|z|^2 > 256$, with a default live budget of 220 iterations.

3.2 Music-dependent equation coefficient

The coefficient $\mu(t)$ is a complex, time-varying perturbation. $B(t)$, $M(t)$, and $Q(t)$ denote the smoothed bass, mid-band, and beat features. E is the equation-modulation control. The exact v0.27 expression is:

EQUATION 2 - AUDIO PERTURBATION

$$\mu(t) = E \{ [(0.18 B(t) + 0.14 Q(t)) \cos(0.43 t)] + i [0.18 M(t) \sin(0.59 t)] \}$$

Bass and detected beats primarily push the real component, while mids push the imaginary component. The two low angular frequencies prevent the perturbation from becoming a direct copy of instantaneous amplitude. Instead, sound changes the direction and magnitude of a slowly rotating coefficient. Because μ multiplies z^2 inside the recurrence, the perturbation can alter the set boundary and internal orbit behavior.

3.3 Equation-level folding and nonlinear warp

Folding is optional and is disabled for the pure recurrence. When enabled, the current iterate is rotated into a moving fold space, blended toward a mirrored coordinate, warped by a bounded quadratic map, and rotated back before the quartic recurrence is evaluated.

EQUATION 3 - ROTATING MIRRORED FOLD

$$q = R(\alpha) z; \quad q_f = (1 - F) q + F (|q| - o(\alpha)); \quad z_f = R(-\alpha)[q_f + 0.32 W S N(q_f)]$$

EQUATION 4 - BOUNDED QUADRATIC WARP

$$N(x,y) = (x^2 - y^2, 2xy) / (1 + x^2 + y^2)$$

Here F is fold amount, W is warp amount, and S is warp scale. The phase is $\alpha = t \omega_f (0.62 + 0.25 M A_f)$. The audio gain is $g = 1 + A_f(0.24 B + 0.17 M + 0.13 Q)$, after which F and W are multiplied by g and clamped. The moving origin is $o = A_o(0.30 + 0.045 \sin(1.17 \alpha), 0.21 + 0.035 \cos(0.91 \alpha))$. Absolute value creates reflection symmetry; the normalized quadratic term adds a smooth deformation without allowing unbounded coordinate growth at large radius.

This operator creates a hybrid dynamical system. It is deliberately identified as an artistic equation fold, not as the unmodified mathematical Mandelbrot set.

4 Three-Dimensional Power-4 Mandelbulb

4.1 Spherical recurrence

A canonical multiplication with all properties of complex multiplication does not exist in three real dimensions. Quartic Pulse therefore uses the White-Nylander spherical-power construction commonly called the Mandelbulb. For point p , initialize $z_0 = p$ inside the distance estimator. At each iteration, compute radius r , polar angle θ , and azimuth ϕ , multiply both angles by the exponent n , raise the radius to n , and add p .

EQUATION 5 - SPHERICAL POWER MAP

$$P_n(r, \theta, \phi) = r^n [\sin(n \theta) \cos(n \phi), \sin(n \theta) \sin(n \phi), \cos(n \theta)]$$

EQUATION 6 - MANDELBULB RECURRENCE

$$z_{(k+1)} = P_n(z_k) + p$$

The Quartic Core preset sets $n_0 = 4$, fold = 0, live detail = 0.55, and audio deformation $A_{3D} = 0.65$. Music changes the exponent according to:

EQUATION 7 - AUDIO-DRIVEN 3D EXPONENT

$$n(t) = \text{clamp}[n_0 + A_{3D}(0.62 B + 0.34 M + 0.48 Q), 2, 10]$$

At the default settings the silent shape is genuinely power 4. Audio raises the exponent rather than replacing it. An optional hybrid fold uses $f(t) = \text{clamp}[f_0 + A_{3D}(0.18 M + 0.12 Q), 0, 1]$ and blends each new iterate toward $|z| - (0.16 + 0.10 \sin(0.21t))$ with weight $0.32 f(t)$. A zero fold preserves the pure spherical recurrence.

4.2 Distance estimate

Direct voxelization would be too expensive for live use. The shader tracks the radial derivative while iterating. If d_r begins at 1, the derivative and distance estimate are:

EQUATION 8 - DERIVATIVE UPDATE

$$d_r \leftarrow n r^{(n-1)} d_r + 1$$

EQUATION 9 - IMPLEMENTED DISTANCE ESTIMATE

$$D(p) = 0.5 \ln(r) r / d_r$$

Iteration stops when $r > 3.2$ or the detail budget is reached. Detail maps to $5 + \text{floor}(6D_l)$ recurrence iterations, producing 5 through 11 iterations for D_l in $[0,1]$. The default $D_l = 0.55$ evaluates 8 recurrence steps.

4.3 Sphere tracing, camera, and shading

For every pixel, a ray starts at an orbiting camera and advances by $\max(0.78 D, 0.00055)$. A surface hit occurs when D is below $0.00075(1 + 0.12s)$, where s is traveled distance. Marching terminates at $s > 8$ or when the adaptive step budget is exhausted. This is a conservative distance-guided ray-marching method related to sphere tracing.

Bass and beats move the camera slightly inward: $d(t) = d_0[1 - 0.055 A_{3D} B - 0.035 A_{3D} Q]$. Mids add a small pitch and orbit oscillation. Surface normals are central finite differences with epsilon 0.0018. Four distance samples along the normal approximate ambient occlusion. Final shading combines key and fill diffuse light, a power-2.4 rim term, a high-frequency-dependent specular exponent, palette bands, atmospheric fog, and a near-surface aura.

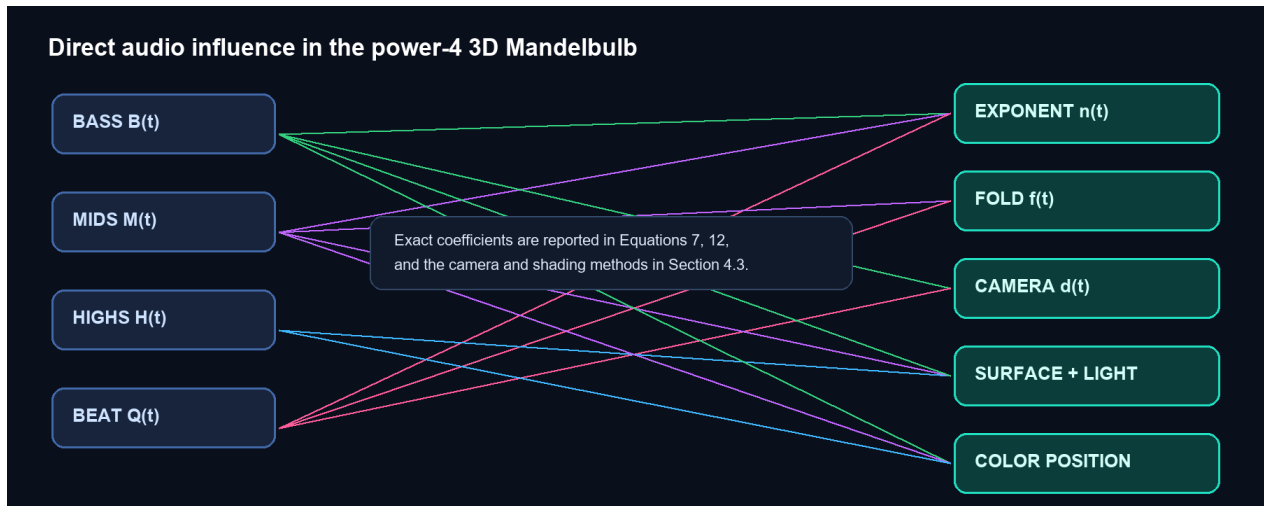


Figure 2. Direct audio influence in the 3D distance-estimated visual. User modulation routes can add further contributions.

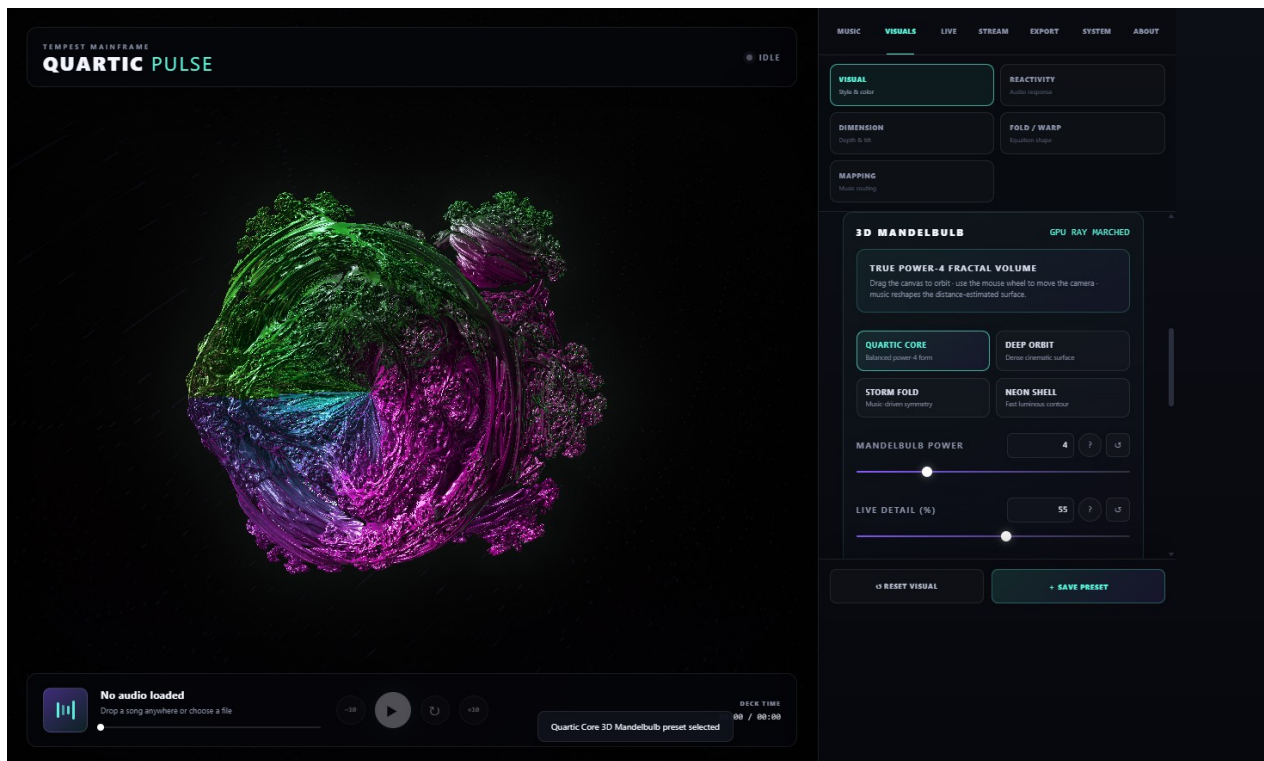


Figure 3. Quartic Core 3D Mandelbulb in the v0.27 interface. The control panel exposes power, detail, deformation, orbit, fold, glow, and camera distance.

5 Audio Feature Extraction

5.1 Sampling and FFT

The analysis context requests a 48,000 Hz sample rate. An `AnalyserNode` uses an FFT size $N = 2048$, producing 1024 nonredundant frequency bins. At 48 kHz, the nominal bin spacing is $48,000/2048 = 23.4375$ Hz. Magnitudes are converted by the browser to unsigned bytes over -90 to -20 dB. The `AnalyserNode` smoothing constant defaults to 0.72.

RMS is computed from 2048 time-domain byte samples after mapping each sample x from $[0,255]$ to $x_n = (x - 128)/128$. The program also samples 64 logarithmically spaced frequency intervals from 20 Hz to $\min(20 \text{ kHz}, \text{Nyquist})$ for conventional spectrum displays.

5.2 Band energies and normalization

In Basic mode the frequency boundaries are 25, 180, 2400, and 12,000 Hz. Advanced mode lets the user move all four boundaries while maintaining valid ordering. The mean byte magnitude in each interval is divided by 255 and multiplied by a visual compensation factor.

EQUATION 10 - RAW BAND FEATURES

$$B_{\text{raw}} = 1.60 \text{ mean}[25,180]; \quad M_{\text{raw}} = 1.45 \text{ mean}[180,2400]; \quad H_{\text{raw}} = 1.80 \text{ mean}[2400,12000]$$

These are visual control signals, not calibrated sound-pressure or loudness measurements. The unequal factors compensate artistically for typical spectral balance and byte-scaled analyzer output.

Table 2. Audio-analysis constants.

Parameter	v0.27 default	Purpose
Sample rate	48,000 Hz requested	Web Audio analysis graph
FFT size	2048 samples	1024 frequency bins
Magnitude range	-90 to -20 dB	Byte conversion range
Analyser smoothing	0.72	Steadier spectrum and color response
Low band	25-180 Hz x 1.60	Bass, camera, exponent, beat
Mid band	180-2400 Hz x 1.45	Imaginary perturbation, fold, orbit
High band	2400-12000 Hz x 1.80	Specular detail, stars, color
Auto target	0.74	Peak-normalized reactivity target

5.3 Asymmetric automatic reactivity

To prevent a loud source from holding every control at 100 percent, automatic reactivity estimates a gain that moves the largest raw band toward 0.74. If G_m is manual sensitivity and P is the largest raw band:

EQUATION 11 - DESIRED AUTOMATIC GAIN

$$G^* = \text{clamp}[0.74 / \max(0.03, P G_m), 0.25, 3.0]$$

The current gain moves 18 percent toward G^* when gain must fall, but only 1.5 percent when it must rise. Fast attenuation limits clipping; slow upward recovery reduces pumping during quiet passages. With automatic mode enabled, normalized features are capped at 0.96. Final bass, mids, highs, and RMS use one-pole smoothing coefficients 0.24, 0.18, 0.20, and 0.20 respectively.

5.4 Beat and frequency-color estimators

A bass baseline follows $A_b \leftarrow 0.975 A_b + 0.025 B$. A beat candidate occurs when $B > 1.34 A_b$ and $B > 0.18$. The detector rearms when $B < 1.12 A_b$ or $B < 0.12$. Accepted beat strength is $Q = \min(1, 1.3B)$, followed by a per-frame decay of 0.84. This lightweight detector favors strong low-frequency onsets; it is not a full tempo or spectral-flux model.

EQUATION 12 - DOMINANT-FREQUENCY COLOR COORDINATE

$$h^* = [0.06 B + 0.45 M + 0.88 H] / [B + M + H]$$

When total energy exceeds 0.035, the palette coordinate moves toward h^* . Low frequencies therefore remain near the start of the palette, mids occupy its center, and highs approach its far end. The response is $0.04 + 0.28(1 - S_a)$, where S_a is the analysis-smoothing control.

6 General Modulation Matrix

Direct formulas provide the signature response, but users may add as many as eight routes from bass, mids, highs, beat, or overall RMS to mathematical and visual targets. Each route has a floor, ceiling, signed amount, attack, and release. For raw source x , floor l , and ceiling u :

EQUATION 13 - ROUTE NORMALIZATION

$$x_n = \text{clamp}[(x - l)/(u - l), 0, 1]$$

EQUATION 14 - ATTACK/RELEASE SMOOTHING

$$\tau = 0.015 + 0.8(1 - v)^2; \quad a = 1 - \exp(-\Delta t / \tau); \quad y \leftarrow y + a(x_n - y)$$

v is attack speed while x_n is rising and release speed while it is falling.

EQUATION 15 - TARGET CONTRIBUTION

$$\Delta p = y (\text{amount}/100) s_{\text{target}}$$

Table 3. Modulation-target scales before final safety clamping.

Target	Scale s_{target}	Resulting control
Equation modulation	0.90	Complex coefficient μ
Equation fold	0.75	Mirrored blend amount
Equation warp	1.00	Bounded nonlinear warp
Dimensional tilt / slice	0.70	Virtual 2D depth transform
Camera zoom	0.45	View magnification

Target	Scale s_{target}	Resulting control
Rotation	0.80	Image-space angle
Color position / flow	0.55	Palette coordinate and motion
Visual motion	1.00	Global motion strength
Pulse jaggedness	0.90	Conventional pulse contour
3D fractal power	1.50	Mandelbulb exponent
3D fold	0.70	Hybrid recurrence fold
3D surface glow	1.00	Rim, specular, and aura

The built-in Math preset routes bass to equation modulation (45 percent), mids to equation warp (30 percent), highs to color position (35 percent), and beat to zoom (48 percent). Users may reverse a route with a negative amount.

7 Rendering and Performance Control

7.1 Two-dimensional rendering

Each pixel iterates the selected recurrence until escape or the iteration cap. Escaped points use a continuous iteration estimate, palette lookup, radial orbit traps, axis traps, boundary emphasis, and audio-dependent brightness. Non-escaped points retain low-level animated interior structure so the set does not become a static black silhouette.

7.2 Adaptive live quality

Live quality responds to measured frame time. If frame time exceeds 23 ms, render scale falls at 0.16 units per second to a minimum of 0.62. Below 18 ms it rises at 0.07 units per second toward 1.0. This hysteresis avoids rapid oscillation near a single threshold.

EQUATION 16 - LIVE MANDELBULB MARCH BUDGET

$$N_{\text{live}} = \text{clamp}[(42 + 54 D_{\text{I}})(0.74 + 0.26 S_{\text{r}}) B_{\text{h}} U, 36, N_{\text{max}}]$$

D_{I} is user detail, S_{r} is adaptive render scale, B_{h} is the hardware budget (0.72 Efficient, 0.88 Balanced, 1.00 Performance), and U is 1.24 only in Unleashed mode. Standard live rendering caps N_{max} at 112; Unleashed or export can reach 192. At the Quartic Core default on Balanced hardware, the initial budget is about 63 ray steps.

7.3 Offline export

Offline export is the default because it can render audio-synchronized frames without maintaining display-rate interaction. The march budget is multiplied by $1.12 + 0.32 \min(2.5, D_{\text{e}})$, where D_{e} is export detail. Export resolution, frame rate, and detail can therefore exceed the live preview while retaining the same audio-feature logic.

8 Engineering Validation

The v0.27 release candidate was validated with automated launch and shader smoke tests on the development system. The test loaded the packaged application, selected the 3D Mandelbulb, injected a synthetic 48 kHz track containing periodic 72 Hz, 760 Hz, and 5.2 kHz bursts, and verified WebGL2 rendering, audio response, controls, presets, and layout.

Table 4. v0.27 engineering smoke-test observations.

Measurement	Observed result	Interpretation
GPU / CPU class	NVIDIA GeForce RTX 3080; 12th Gen Intel Core i9-12900K	Development workstation
Packaged canvas	944 x 881 pixels	Live application stage during smoke test
Frame rate	58 FPS packaged; 60 FPS development	Near display-rate operation
Frame time	17.4 ms packaged; 16.7 ms development	Below adaptive downscale threshold
Adaptive scale	100 percent	No resolution reduction on test system
Feature checks	WebGL2, 3D controls, presets, audio response passed	Functional validation

These results demonstrate successful execution on one high-end system; they are not a population benchmark. Performance on integrated GPUs and lower-power systems remains dependent on resolution, driver, thermal limits, browser GPU backend, active audio source, and selected detail.

9 Discussion and Limitations

Mathematical status. The two-dimensional zero-fold quartic mode is an escape-time rendering of a power-4 complex recurrence with an explicitly documented time-dependent coefficient. Nonzero folding produces a hybrid map. The 3D mode is a spherical-power Mandelbulb construction and not a unique three-dimensional extension of complex arithmetic.

Signal status. Band values are normalized byte magnitudes and are not calibrated dB SPL, LUFS, or psychoacoustic loudness. Simple band means can be influenced by windowing, leakage, source equalization, and sample-rate differences. The beat detector is intentionally lightweight and bass-oriented.

Numerical status. The Mandelbulb distance estimator, finite-difference normal, occlusion samples, and ray hit threshold are approximations chosen for stable real-time images. Equation folding can weaken the assumptions behind a strict distance bound, so the 0.78 march factor is intentionally conservative but cannot guarantee artifact-free tracing for every preset.

Temporal status. Several analysis smoothing coefficients are applied once per rendered frame. Extremely variable frame rates may therefore change effective response time. The configurable modulation matrix uses Delta-t-correct exponential smoothing and is more frame-rate invariant.

Future evaluation should benchmark multiple GPU tiers, convert all temporal filters to explicit time constants, compare spectral-flux onset detection with the present beat method, add perceptual loudness normalization, and quantify offline frame determinism across repeated exports.

10 Reproducibility

The implementation is contained in `src/renderer/app.js` in the Quartic Pulse repository. Reproduction requires a Windows 11 system with WebGL2-capable graphics, Node.js-compatible project tooling, and the project dependencies. The GPL license permits inspection, modification, and redistribution of the source and derivative versions under its terms.

Recommended reproduction sequence: run the development application; select Visuals > 3D Mandelbulb > Quartic Core; leave fold at zero; load a test track; record B, M, H, Q, frame time, render scale, and the selected controls; then repeat with audio deformation set to zero. The difference isolates direct music-to-equation coupling from idle camera motion and palette flow.

Table 5. Reproduction defaults for the signature 3D preset.

Quartic Core parameter	Default	Meaning
Mandelbulb power <code>n_0</code>	4.0	Silent spherical exponent
Live detail <code>D_l</code>	0.55	8 recurrence steps; base ray budget 71.7 before hardware scaling
Audio deformation <code>A_3D</code>	0.65	Gain on direct audio-to-3D formulas
Auto orbit	0.28	Slow signed camera orbit
Hybrid fold <code>f_0</code>	0.00	Pure Mandelbulb recurrence
Surface glow	1.20	Rim, specular, and aura intensity
Camera distance <code>d_0</code>	3.75	Silent camera radius
Hardware budget	0.88 Balanced	Average-PC ray-step multiplier

11 Conclusion

Quartic Pulse implements a direct bridge from music analysis to fractal mathematics. In the two-dimensional mode, sound changes a coefficient inside a quartic complex recurrence and can control a documented fold-and-warp operator. In the three-dimensional mode, sound changes the spherical exponent, hybrid fold, camera, surface bands, and lighting of a distance-estimated power-4 Mandelbulb. Automatic gain, bounded controls, attack/release smoothing, adaptive resolution, and tiered ray budgets make this coupling usable in a live Windows 11 application. The defining idea is therefore precise: the music is not only driving the shader's appearance; it is driving parameters of the mathematics that the shader evaluates.

A Appendix A - Consolidated Formula Sheet

Table A1. Key equations transcribed from v0.27.

Subsystem	Implemented expression
2D quartic	$z_{(k+1)} = z_k^4 + \mu(t) z_k^2 + c'$
2D audio coefficient	$\mu(t) = E\{[(.18B+.14Q)\cos(.43t)] + i[.18M \sin(.59t)]\}$

Subsystem	Implemented expression
Fold gain	$g = 1 + A_f(.24B + .17M + .13Q)$
Quadratic warp	$N(x,y) = (x^2 - y^2, 2xy) / (1 + x^2 + y^2)$
3D exponent	$n(t) = \text{clamp}[n_0 + A_3D(.62B + .34M + .48Q), 2, 10]$
3D fold	$f(t) = \text{clamp}[f_0 + A_3D(.18M + .12Q), 0, 1]$
Distance estimate	$D = 0.5 \ln(r) \ r / d_r$
Auto reactivity	$G^* = \text{clamp} [.74 / \max(.03, P \ G_m), .25, 3]$
Beat	$B > 1.34 \ A_b$ and $B > .18$; rearm below $1.12 \ A_b$ or $.12$
Color coordinate	$h^* = (.06B + .45M + .88H) / (B + M + H)$
Route response	$\tau = .015 + .8(1 - v)^2$; $a = 1 - \exp(-\Delta t / \tau)$
Live ray budget	$(42 + 54D_l) (.74 + .26S_r) B_hU$, clamped to 36-112/192

B Appendix B - 3D Preset Values

Table B1. Shipped 3D presets. UI percentages reflect each control's displayed scale.

Preset	Power	Detail	Audio	Orbit	Fold	Glow	Camera
Quartic Core	4.00	55%	65%	+14% UI	0%	60% UI	50% UI
Deep Orbit	4.00	84%	48%	+8% UI	8%	72.5% UI	33% UI
Storm Fold	4.35	70%	100%	+21% UI	72%	75% UI	53% UI
Neon Shell	6.00	45%	78%	-17.5% UI	28%	100% UI	62% UI

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